WHY PYBIO-10

From the bench to the code.

Spreadsheets break when the data gets big. In 10 evenings you learn to clean, analyse, visualise and model your own biological data in Python, the language used across modern genomics, bioinformatics and lab analytics.

```python
# Day 7: your first sequence scan
from Bio import SeqIO
from Bio.SeqUtils import gc_fraction
for r in SeqIO.parse("genes.fasta", "fasta"):
    print(r.id, gc_fraction(r.seq))
```

By Day 10 you will be able to

- Write clean, reusable Python in Jupyter and Google Colab
- Turn messy lab and field data into tidy, analysis-ready tables
- Run t-tests, ANOVA and regression, and make publication-ready figures
- Read FASTA and GenBank files, run BLAST, and handle gene-expression data
- Build and evaluate your first machine-learning model for biology

AVYAGRAHA
RESEARCH AND ANALYTICS LLP

AVYAGRAHA RESEARCH AND ANALYTICS LLP PRESENTS

REGISTRATIONS OPEN

PyBio-10

Python for Life Sciences

10-DAY HANDS-ON TRAINING PROGRAMME

DATES
16 – 26 Nov 2026
Sunday 22 Nov holiday

TIMING
6:30 – 8:30 PM IST
2 hours per evening

MODE
Online · Live
Hands-on, no prior coding

FEE (INCL. GST)
₹1,499 / ₹1,999
Students / Professionals

RECORDINGS
3-month access
Revisit every session

CERTIFICATE
E-certificate
Plus capstone projects

BONUS 20+ AI tools for research

Python · NumPy · Pandas · Matplotlib · Seaborn · SciPy · Biopython · scikit-learn

WHO CAN REGISTER

Built for life scientists, not programmers.

- UG, PG and PhD students in the life sciences
- Faculty, scientists and research associates
- Biotechnology, Microbiology, Biochemistry, Genetics, Agriculture, Pharmacy and Medical Sciences
- Pharma, agri-biotech and healthcare professionals
- Anyone moving from Excel or SPSS to code-based analysis

No prior programming experience needed. A laptop with internet access and a Google account (for Colab) is all you need.

10	20 h	11
live evening sessions	of hands-on coding	tools and libraries

Start where you are.
Code your own data.

ENROLL NOW

aviqlabs.com/pybio10

Students
UG / PG / PhD students and research scholars
₹1,499

Professionals
Faculty, scientists, industry and others
₹1,999

All fees inclusive of GST · Limited seats

1. **Scan the QR** or visit aviqlabs.com/pybio10
2. Tap **Pay now**: your UPI app opens with the fee filled in
3. Note the **UPI transaction ID** after paying
4. Enter it, **upload the payment screenshot** and submit

Questions?

Call or WhatsApp
+91 83105 79116 · +91 94803 63456
Email **avinashstat@aviqlabs.com**

COURSE STRUCTURE

10 days. 10 skills.

Every session is live and hands-on: you write and run the code yourself, on biological datasets.

Module 1 · Python Foundry Days 1–2

D01 Inauguration · Python foundations
Colab notebooks, variables, data types, operators
Mon 16 Nov

D02 Thinking in code
Data structures, conditions, loops, functions
Tue 17 Nov

Module 2 · Scientific Data Toolkit Days 3–5

D03 NumPy for scientific computing
Arrays, broadcasting, vectorised calculations
Wed 18 Nov

D04 Pandas for lab data
Clean, validate, merge and reshape datasets
Thu 19 Nov

D05 Publication-ready visualisation
Matplotlib, Seaborn, multi-panel figures
Fri 20 Nov

Module 3 · Statistics and Sequences Days 6–7

D06 Biostatistics in Python
t-tests, ANOVA, regression, assumptions
Sat 21 Nov

D07 Biopython: sequence analysis
FASTA, GenBank, translation, BLAST
Mon 23 Nov

Module 4 · Omics and ML Days 8–9

D08 Omics data handling
Expression matrices, FDR, volcano plots
Tue 24 Nov

D09 Machine learning for biology
Classification, evaluation, clustering, PCA
Wed 25 Nov

Module 5 · Capstone Day 10

D10 Capstone and valedictory
End-to-end analysis, presentations, e-certificates
Thu 26 Nov

Sunday 22 November — HOLIDAY · NO SESSION — Every session: 6:30 – 8:30 PM IST

TOOLS YOU WILL USE: `Python 3` `Jupyter` `Google Colab` `NumPy` `Pandas` `Matplotlib` `Seaborn` `SciPy` `statsmodels` `Biopython` `scikit-learn`

COORDINATOR AND RESOURCE PERSONS

Learn from practitioners.



COORDINATOR · DAYS 1, 6–10

Dr. Avinash Gopa

Founder & CEO, Avyagraha Research & Analytics LLP · Ex-ICMR Project Scientist-II · Ph.D. (Agricultural Statistics), IARI, New Delhi



DAYS 1, 4 AND 5

Mr. Chandan G Y

Data Analyst, Bangalore



DAYS 2 AND 3

Mr. Rajesh A

CEO & Tech Member, Avinya Techsphere, Bangalore



DAY 6

Mr. Shrey Pandya

Research Analyst, Population Research Centre, Department of Statistics, The Maharaja Sayajirao University of Baroda, Vadodara



DAY 9

Dr. Sandip Garai

Scientist, ICAR-Indian Institute of Agricultural Biotechnology, Garhkhatanga, Ranchi

KEY TAKEAWAYS

You walk away with

.py — Code notebooks
Every day's notebook, ready to reuse on your own research data

3M — 3-month recordings
Revisit any session at your own pace

CP — Capstone projects
End-to-end analyses you can show on your CV

EC — E-certificate
Issued on completion of the programme

DNA — Real biological datasets
Sequence, expression, lab and field data from published studies

Q&A — Live doubt clearing
Direct interaction with the trainers in every session

AI — Bonus: 20+ AI tools for research
A curated introduction to AI tools that support research work

ABOUT AVYAGRAHA

AVYAGRAHA
RESEARCH AND ANALYTICS LLP

A DPIIT-recognised deep-tech startup, incubated at AIC-UAS-B, GKVK, Bengaluru. We provide data-driven solutions, training programmes and research collaborations.

- **Training:** statistics, R, Python, AI/ML and Generative AI for institutions and professionals
- **Research consultancy:** study design, data analysis and statistical modelling
- **Partnerships:** MoUs with universities and R&D institutes

Register now

Scan the QR, pay by UPI and upload your payment screenshot.

`aviqlabs.com/pybio10`

Contact us

+91 83105 79116 · +91 94803 63456

avinashstat@aviqlabs.com

www.aviqlabs.com

DETAILED TRAINING PROGRAMME CURRICULUM

PyBio-10: Python for Life Sciences
10-Day Hands-On Training Programme

16 Nov 2026 – 26 Nov 2026 · 6:30 PM to 8:30 PM IST · Online (live) · Sunday holiday

Organized by Avyagraha Research and Analytics LLP, Ramasagara, Ballari, Karnataka

www.aviqlabs.com · avinashstat@aviqlabs.com · +91 83105 79116 · +91 94803 63456

10 live sessions | 20 hours hands-on | 3-month recordings | Capstone projects | E-certificate

Date	Topics	Resource person
Module 1 · Python Foundry (Days 1–2)		
16/11/2026 Monday	D01 **Inauguration and Python foundations** Programme orientation and setup; notebooks, cells and Markdown; variables, naming, numbers, strings and Boolean values; operators, type conversion, print(), input() and simple errors. Practical: a documented notebook of sample IDs, treatment labels and measurements with a simple summary.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha & **Mr. Chandan G Y** Data Analyst, Bangalore
17/11/2026 Tuesday	D02 **Thinking in code: data structures and logic** Lists, tuples, dictionaries and sets; if / elif / else conditions; for and while loops over samples; functions, parameters, return values and basic debugging. Practical: count DNA bases or codons, flag out-of-range measurements and convert assay units with a reusable function.	**Mr. Rajesh A** CEO & Tech Member, Avinya Techsphere, Bangalore
Module 2 · Scientific Data Toolkit (Days 3–5)		
18/11/2026 Wednesday	D03 **NumPy for scientific computing** Arrays, shape, dimensions, data types and indexing; slicing, reshaping and broadcasting; vectorised arithmetic and axis-wise aggregation; random numbers, a simple simulation and seeds for reproducibility. Practical: process a plate-reader absorbance matrix with means, SDs and blank correction.	**Mr. Rajesh A** CEO & Tech Member, Avinya Techsphere, Bangalore
19/11/2026 Thursday	D04 **Pandas for biological and laboratory data** Series and DataFrames; reading CSV/Excel; data types, duplicate records, missing values and basic validation; filtering, sorting, grouping and summary tables; merge, groupby, pivot and melt for tidy data. Practical: clean a lab dataset, join it with sample metadata and keep a data-quality log.	**Mr. Chandan G Y** Data Analyst, Bangalore
20/11/2026 Friday	D05 **Publication-ready scientific visualisation** Figure, axes, labels, legends, scales and annotations; box, violin and scatter plots, regression lines and heatmaps; error bars and communicating variation; multi-panel layouts and export to high-resolution PNG, TIFF or PDF. Practical: two labelled figures for a treatment-versus-control dataset.	**Mr. Chandan G Y** Data Analyst, Bangalore
Module 3 · Statistics and Sequences (Days 6–7)		
21/11/2026 Saturday	D06 **Biostatistics in Python** Descriptive statistics and exploratory checks; normality, variance and study-design assumptions; independent/paired t-tests and chi-square tests; one-way/two-way ANOVA with Tukey post-hoc; correlation and linear regression with SciPy and statsmodels. Practical: a short statistical report on a treatment-control dataset.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha & **Mr. Shrey Pandya** Research Analyst, Dept. of Statistics, MSU Baroda, Vadodara
22/11/2026 Sunday	HOLIDAY	

Date	Topics	Resource person
23/11/2026 Monday	D07 **Biopython essentials: sequence analysis** Seq objects and DNA, RNA and protein basics; GC fraction, transcription and translation; parsing FASTA and GenBank with SeqIO; NCBI Entrez and responsible use of public records; pairwise alignment and BLAST output. Practical: a reusable sequence-analysis notebook.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha
Module 4 · Omics and Machine Learning (Days 8–9)		
24/11/2026 Tuesday	D08 **Omics data handling** Gene-expression matrices and sample metadata; FASTQ, SAM/BAM and VCF formats; counts versus normalised expression and log transformation; fold change, p-values and false discovery rate (FDR); volcano plots and clustered heatmaps. Practical: explore a prepared expression dataset and visualise results.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha
25/11/2026 Wednesday	D09 **Machine learning for biology** Features, target, train/test split and preprocessing; logistic regression and random forest; confusion matrix, precision, recall, ROC curve and class imbalance; k-means and hierarchical clustering; PCA, data leakage and overfitting. Practical: train and evaluate a baseline classifier on held-out data.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha & **Dr. Sandip Garai** Scientist, ICAR-IIAB, Ranchi
Module 5 · Capstone (Day 10)		
26/11/2026 Thursday	D10 **Capstone, certification and valedictory** Capstone workflow: question, data, quality checks, analysis and visualisation; reproducibility with clear cells, comments and fixed seeds; short presentations, feedback and valedictory. Output: a runnable notebook, a labelled figure and a brief summary with limitations.	**Dr. Avinash Gopa** Founder & CEO, Avyagraha

Programme fees

Category	Fee (incl. GST)
Students: UG / PG / PhD and research scholars	**Rs. 1,499 /-**
Professionals: faculty, scientists, industry, others	**Rs. 1,999 /-**

Requirements

- A laptop or desktop with a stable internet connection.
- A Google account, for Google Colab notebooks.
- No software to install: everything runs in the browser.

How to register

1. Scan the QR code or visit **aviqlabs.com/pybio10**.
2. Tap **Pay now** (any UPI app opens with the fee filled in), or pay to UPI ID **9480363456@okbizaxis** (Avyagraha Research and Analytics LLP).
3. Note the **UPI transaction ID** shown after payment.
4. Enter the transaction ID, **upload the payment screenshot** and submit to get your registration ID by email.
5. Joining details will be shared by email and WhatsApp before the programme begins.

aviqlabs.com/pybio10